

Supplement to “Simulation-based comparison of offline value learning and online mixed-integer control for battery dispatch”

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1. Evaluation protocol and numerical checks

The benchmark uses IT_NORD data from 2016–2017 for training, 2018 for validation, and 2019 for test. All controllers share the economic objective, 15-minute update interval, feasible action set, and terminal treatment. Checkpoint and restart selection use validation cost alone; batch selection starts from the feasible zero-action policy. All action-grid candidates lie within the computed feasible interval before objective evaluation. Evaluation configurations and input/output hashes are documented in the GitHub repository (Section 7).

Reference-value attribution and nested scenario-count sensitivity with $M \in \{2, 8, 16\}$ and three fixed streams are exploratory analyses. They address the contribution of learning and the sensitivity to comparator resolution, separately from the primary comparisons. The high-fee 15-second analyses describe sensitivity to the solver budget; the primary online comparison uses five seconds per solve.

The controller set reported in the main paper is:

1. no storage;
2. validation-tuned price-threshold control;
3. conditional-mean convex-envelope MPC;
4. conditional-mean exact-band mixed-integer quadratic program (MIQP);

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5. reflected two-scenario, two-stage exact-band MIQP with a shared first action;
6. the no-learning reference-value one-step controller; and
7. the complete offline learned procedure: five restarts, historical validation selection against a zero-action feasible incumbent, one-step hard-cost deployment, and the common safety set.

The central fee cell contains five independent five-restart batches. No storage is a scale reference. Comparator scope is kept explicit: the primary online controls are the conditional-mean and scenario-based exact-band MIQPs, and the convex envelope isolates tariff representation. All reported methods use the common evaluation protocol.

The price rule charges below a lower price threshold, discharges above an upper threshold, and otherwise holds. Candidate thresholds use training-price quantiles; the inactive no-action rule is included. The pair is selected by hard common cost on the same 30 validation days used for checkpoint selection.

2. Data processing and calibration

The OPSPD 60-minute release dated 2020-10-06 [1] supplies local load, day-ahead price, solar, and wind. Renewable components are summed with at least one available component. A calendar year is eligible only if all required series have no more than 1% missing rows. Gaps of at most two consecutive hours are interpolated in time; days with a remaining missing value are removed.

The IT_NORD record has four consecutive eligible years and 35,040 cleaned hours. Before interpolation, missing fractions are 0 for load, 2.852×10^{-4} for price, and 2.567×10^{-4} for renewable generation; all are zero afterward. Conversion to Europe/Rome time precedes season and weekday classification. Daylight-saving days with other than 24 rows are excluded from daily evaluation.

Only training data determine the load median, renewable 95th percentile, hourly regime medians, Ornstein–Uhlenbeck (OU) coefficients, correlation, and reflection box. Raw load and renewable shapes are scaled to a 500-kW base microgrid. Prices are converted from EUR/MWh to EUR/kWh. The battery has 1,000 kWh capacity, 500 kW charge and discharge limits, 0.95 one-way efficiencies, and an SoC interval $[0.1, 0.9]$. Initial and target SoC are 0.5. The linear throughput coefficient is $300/(2 \times 4000) = 0.0375$ EUR/kWh. The common terminal coefficient is $\lambda_T = 10,819.79$ EUR per squared SoC

fraction: a 0.10 terminal deviation costs 25% of the median complete-day no-storage bill in training. It is then held fixed across methods. The cycle-based proxy is used for controlled comparison rather than as an electrochemical lifetime model [2].

For each regime, a zero-intercept first-order autoregressive coefficient is fitted by least squares to within-day consecutive residual pairs. The exact discrete-time mapping gives $\kappa = -\log(\phi)$ per hour and maps the innovation variance to σ . The cross-process parameter ρ is set to the correlation of standardized hourly load and price innovations, used as a proxy for instantaneous correlation rather than a joint continuous-time fit. Reflection bounds equal the training residual range widened by 20% on each side.

Table 1: Fixed physical, economic, controller, and computational parameters. Values are calibrated from training data or specified by the common experimental design.

Quantity	Value
Battery energy; charge/discharge limits	1000 kWh; 500/500 kW
SoC bounds; initial/target SoC	[0.10, 0.90]; 0.50/0.50
Charge/discharge efficiency; taper width	0.95/0.95; 0.05 SoC
Horizon; control/simulation interval	24 h; 15/5 min
Base power; renewable scaling	500 kW; $\gamma_R = 1.2$
Export fraction; throughput cost	$\alpha_s = 0.50$; $\lambda_1 = 0.0375$ EUR/kWh
Terminal coefficient; terminal target	$\lambda_T = 10, 819.79$ EUR/SoC ² ; $s_{\text{tar}} = 0.50$
Tariff threshold; fee anchors; smooth widths	300 kW; 20/40/80 EUR/h; 5/10/20 kW
Hard deployment search; quadrature	1,025-point structural grid; 3×3 Gauss-Hermite
Primary online-MIQP budget; target gap	5 s per solve; 1%
Solver feasibility tolerance; binary separation	10^{-9} ; 10^{-5} kW
Scenario sensitivity	$M = 2, 8, 16$; three nested streams; pool size 16
Network; Fourier harmonics	5 layers $\times$ 96; 1, 2, 3
Training per PI round	at most 2,000 Adam and 80 L-BFGS steps
PI rounds; selection margin	4; 0.10 EUR/day

The winter, spring, summer, and autumn weekday calibrations use 2,856, 3,048, 3,072, 3,072 training hours, respectively. For the two outer winter fees,

Table 2: Weekday OU calibration and central-fee neural output scale β by season. Mean-reversion rates are per hour; diffusion scales are in kW/ $\sqrt{h}$ and EUR/kWh/ $\sqrt{h}$; β is in EUR/h.

Season	Load error ($\kappa_y, \sigma_y, [y_{\min}, y_{\max}]$)	Price error ($\kappa_p, \sigma_p, [p_{\min}, p_{\max}]$)	ρ	β (EUR/h)
Winter	(0.0579, 33.28, [-536.6, 533.9])	(0.0394, 0.00580, [-0.0657, 0.1696])	0.132	301.55
Spring	(0.0607, 38.07, [-420.8, 720.7])	(0.0779, 0.00276, [-0.0331, 0.0759])	0.176	295.45
Summer	(0.0288, 24.42, [-565.0, 603.5])	(0.0449, 0.00449, [-0.0461, 0.1279])	0.122	297.98
Autumn	(0.0582, 38.55, [-424.6, 639.9])	(0.0889, 0.00602, [-0.0508, 0.1111])	0.118	298.73

the neural output scales are 300.95 (20 EUR/h) and 302.75 (80 EUR/h). The hard-band binary separation is 10^{-5} kW. State-domain reflection bounds, hourly profiles, and every remaining calibrated coefficient are stored in the machine-readable parameter file.

3. Exact-band MIQP formulations

For each horizon index h , introduce nonnegative charge u_h^c , discharge u_h^d , import g_h^+ , export g_h^- , and binary band variable z_h . The signed action is $a_h = u_h^d - u_h^c$, and

$$s_{h+1} = s_h + \frac{\Delta}{E_{\max}} \left(\eta_c u_h^c - \frac{u_h^d}{\eta_d} \right), \quad (1)$$

$$g_h^+ - g_h^- = n_h - u_h^d + u_h^c. \quad (2)$$

Taper and one-step safe constraints are imposed separately on charge and discharge. Direction binaries $d_h^c, d_h^g \in \{0, 1\}$ impose

$$u_h^c \leq a_c d_h^c, \quad u_h^d \leq a_d (1 - d_h^c), \quad (3)$$

$$g_h^+ \leq \bar{g}_h^+ d_h^g, \quad g_h^- \leq \bar{g}_h^- (1 - d_h^g). \quad (4)$$

They are present at every price: dominance of simultaneous flows at positive prices is not sufficient for arbitrary time-limited incumbents. For valid bounds $\underline{g}_h \leq g_h \leq \bar{g}_h$, the band logic uses

$$g_h \leq G_{\text{thr}} + (\bar{g}_h - G_{\text{thr}})z_h, \quad (5)$$

$$g_h \geq G_{\text{thr}} + \epsilon - (G_{\text{thr}} + \epsilon - \underline{g}_h)(1 - z_h). \quad (6)$$

Here $\epsilon = 10^{-5}$ kW and SCIP’s feasibility tolerance is 10^{-9} . The separation specifies a numerical representation of the strict band boundary, not a tolerance in the realized bill. Before holding the first action, native clipping enforces the common safe interval. If the incumbent has $z_0 = 0$ but its recovered action exceeds the band by at most ϵ , the action is moved to a verified fee-inactive crossing only when that crossing is feasible and the change is at most ϵ . Otherwise no band recovery is applied. Raw and held actions, direction flows, band status, and the recovery size are retained in each solve record. Every realized bill uses $\mathbf{1}\{N - a > G_{\text{thr}}\}$ without a tolerance. The objective is

$$\sum_h \Delta \left[c_h g_h^+ - \alpha_s c_h g_h^- + \lambda_1 (u_h^c + u_h^d) + c_f z_h \right] + \lambda_T (s_H - s_{\text{tar}})^2. \quad (7)$$

Thus the terminal treatment is identical to that of the learned controller. An epigraph constraint passes the quadratic terminal term to SCIP [3]. A hard terminal equality is used only in an ablation.

The stochastic MIQP replicates these variables by scenario. Charge and discharge at $h = 0$ are equal across scenarios. Later variables are scenario-contingent and the objective is the arithmetic scenario mean. Exact OU transition variances are used. When the two mean-reversion rates differ, the innovation covariance over one interval is

$$\rho \sigma_y \sigma_p \frac{1 - \exp\{-(\kappa_y + \kappa_p)\Delta\}}{\kappa_y + \kappa_p}. \quad (8)$$

Each simulated error is reflected into its training box. A no-storage primal start supplies an incumbent before the online solve. The scenario-count analysis draws one pool of 16 paths per stream, date, and decision epoch, then uses its first M paths. The samples are therefore nested across $M = 2, 8, 16$. Streams 7301, 7302, and 7303 are all reported under the same five-second budget and one-thread SCIP setting.

4. Learned value and deployment details

OpenAI Codex (OpenAI; accessed September 2026) assisted with code review and experiment and Matplotlib visualization scripting. It was not a controller or a source of simulated or historical observations. The author reviewed the scripts and figures, verified references and numerical outputs, and retains responsibility for the research design and reported results.

The independent decision gate is a development-region winter problem with state (S, Y) , deterministic price error $P = 0$, and $c_f = 0$. It keeps the

24-hour horizon and battery model but is not evidence about the band-fee mechanism. Monotone finite-difference policy iteration is run on $21 \times 21 \times 25$ and $31 \times 31 \times 37$ SoC–error–time grids; neural value and action metrics use 5,000 fixed Sobol audit states. Threshold-specific correctness is tested separately by the hard one-step search, dense-grid comparison, and realized threshold-occupation diagnostics.

For the selected four-PI gate checkpoint, the fraction of audit actions with absolute error above 10% of rated power (50 kW) was 35.2%. Mean and 95th-percentile local Hamiltonian regret on the uniform audit design were 6.14 and 17.70 EUR/h; the maximum was 597.53 EUR/h. These diagnostics complement the aggregate policy-cost comparison in the main paper without implying uniform decision accuracy.

The network has five hidden layers of width 96, hyperbolic tangent activation, and time Fourier features at harmonics 1, 2, and 3. Inputs are affinely mapped to $[-1, 1]$, and all PDE derivatives use float64 automatic differentiation. The correction network is trained for at most 2,000 Adam steps in each of four policy-iteration rounds, followed by at most 80 L-BFGS iterations. Training, adaptive candidates, tuning validation, final audit, and test simulation have separate fixed seeds. Residual-adaptive refinement is disabled in the benchmark configuration because the decision gate did not justify its added complexity.

For a fixed policy, the loss sums the mean squared PDE residual divided by β^2 and unit-weight Neumann penalties on the exogenous-state faces. The latter square $(y_{\max} - y_{\min})\partial_y v/B$ and $(p_{\max} - p_{\min})\partial_p v/B$ on their respective faces, where $B = B_0 + \lambda_T \max_{s \in [s_{\min}, s_{\max}]} (s - s_{\text{tar}})^2$ and B_0 is the training-period no-storage daily-cost scale. SoC admissibility is handled by the action set, and the ansatz enforces the terminal value. Adam uses 2,048 interior and 256 boundary points per step; L-BFGS uses fixed designs of 8,192 and 2,048 points. Within a policy-evaluation round, synthetic residual RMS selects the Adam iterate and rejects a worsening L-BFGS update. Historical economic validation instead selects among completed PI checkpoints and restarts.

The reference-policy term integrates the no-storage smooth cost on the OU conditional-mean path by 24-point midpoint quadrature. Its final layer is zero initialized. Finite differences in physical time, SoC, load error, and price error check the automatic-differentiation chain rule after adding this term. The interpolated profiles are piecewise linear, so a time-difference stencil may straddle a slope change of the quadrature reference.

For the no-learning counterfactual, the neural correction is set identically to zero before the hard one-step minimization. The no-storage running part of the next reference value is action-independent; after dropping

this common constant, the code minimizes the exact current hard cost plus $\lambda_T(s_{k+1} - s_{\text{tar}})^2$. Tests compare all pairwise candidate-value differences and the minimizing action with a zero-correction network. The counterfactual uses the same 1,025-point structural candidate set and safety map as learned deployment.

Deployment evaluates the exact hard stage cost plus a learned next value on 1,025 equally spaced feasible actions, augmented by both endpoints, zero, the tariff crossing, and verified feasible candidates on each nonempty fee side. An adjacent action value need not change $N - a$, particularly at zero action; candidate construction therefore also uses adjacent grid-exchange values. A 3×3 Gauss–Hermite rule integrates the correlated next error. The last decision uses the exact terminal quadratic. Tests compare the returned decision with a 32,769-point independent grid and separately verify the tariff crossing. Because the feasible interval is at most 1,000 kW wide, the uniform-mesh term in the main paper’s hard-threshold grid-defect result is at most 0.977 kW; the reported denser-grid audit measures the realized objective defect directly. Reported exact-threshold occupation uses $|G - G_{\text{thr}}| \leq 10^{-6}$ kW; fixed 5, 10, and 20 kW neighborhoods are reported separately. The post-selection projection ablation instead minimizes the same one-step objective on $\mathcal{A}_{\text{cont}}(s)$, projects that minimizer to $\mathcal{A}_{\text{dep}}(s)$, and reevaluates the projected action. It uses the same selected weights, quadrature, grid size, days, and common cost as direct deployment and does not feed back into selection.

After batch selection, a finite Bellman audit samples four uniform states at each of the 96 decision indices for every selected trained central-fee batch. It recomputes the one-step minimum on a 4,097-point grid with a 7×7 quadrature reference and reports residual oscillations and the total deployed-action defect. A separate 4,097-point search holds the deployed 3×3 quadrature fixed to isolate grid-only defects. The audit also reports the sampled value change from 3×3 to 7×7 quadrature and terminal error. The design and seed are fixed, but sampled maxima are not treated as continuous-state suprema or uniform quadrature bounds.

5. Selection, inference, and computation

Every PI checkpoint is evaluated on the first 30 eligible same-regime validation weekdays. All checkpoints receive identical days and exact cost accounting. Sequential selection starts at iteration 0 and changes the incumbent only for a mean improvement of at least 0.10 EUR/day. One five-seed batch is then treated as a complete randomized training-and-selection procedure. For the central winter cell, seeds 0–24 form five consecutive five-restart

batches. At batch selection, zero action is the initial feasible incumbent and is retained unless a trained winner lowers validation cost by at least 0.10 EUR/day. Its retention frequency is reported separately; a retained fallback is not attributed to neural control.

Daily differences are paired by date. The primary interval is a one-sided 95% circular moving-block bootstrap [4] with seven-eligible-day blocks and 10,000 replicates. At the central fee, each replicate also resamples the five complete training batches with replacement; the same sampled date blocks are applied to learned and comparator outcomes. Outer-anchor inference is conditional on one complete batch. For each scenario count, the paired bootstrap includes a third randomization level: it resamples the three scenario streams with replacement, the complete learned-policy batches with replacement, and common circular date blocks. The same resampled dates are used on both sides of each learned-minus-MIQP difference. Between-stream standard deviations are also reported without treating three streams as a population estimate. Two-sided intervals and Holm adjustments [5] are secondary. No equivalence claim is based on an interval containing zero, and no noninferiority test is run without an externally justified margin.

Online time starts immediately before the controller call and ends after the held action is returned. It includes forecasting, solver or quadrature work, and action search, but excludes the common post-call safety gate and plant integration. Daily mean, median, and 95th-percentile latencies are recorded; tables average the daily summaries rather than pooling calls. Solver tables report incumbent availability, status, relative gap, node count, and time-limit frequency. Offline accounting sums each restart’s recorded training-run elapsed time, including post-fit diagnostics and certification, and controller-call time for all four checkpoints on all 30 validation days. Training manifests have one-second resolution; their elapsed times replace, rather than supplement, optimizer-only timers. Selected-checkpoint loading is measured separately through device synchronization. Process setup before the training manifest, shared validation simulation and file I/O are outside this accounting. Both high-fee exact-band MIQPs are also evaluated with a 15-second limit in a descriptive solver-budget analysis, separate from the primary five-second deployment comparison.

The representative trajectory date is selected using no-storage output only. Among the 30 winter dates, the procedure computes the median no-storage threshold-exceedance duration, chooses the observed duration closest to that median, and breaks ties by earliest date. All four trajectories—learned, reference, two-scenario MIQP, and 16-scenario MIQP—use this date; the displayed learned policy is the first validation-selected batch. Both

MIQP trajectories replay recorded held actions. Learned and reference trajectories use the reported policies, and all four cost decompositions match their corresponding daily records. Figure-generation time is not controller latency.

Experiments ran on a Windows 10 workstation with an AMD Ryzen 7 9700X (8 cores, 16 logical processors) and an NVIDIA GeForce RTX 4080 SUPER (16,376 MiB). The environment used Python 3.9.19, PyTorch 2.4.1 with CUDA 12.4, PySCIPOpt 6.2.1, and SCIP 10.0. Each mixed-integer solve used one CPU thread. Day-level parallelism ranged from two workers per scenario stream (at most six concurrent solves) to a shared pool of at most eight workers. Scenario and seasonal jobs could overlap with GPU evaluation and short solver diagnostics. Every solve and BLAS backend remained single-threaded. Execution records document the concurrency settings. Learned-value training, validation, and held-out evaluation used the GPU; neural call latency includes transfer of one-step values back to the CPU. These are hardware-configured controller timings, not a CPU-only neural comparison or measurements under identical background load. SCIP numerical-trouble messages are retained in the computational logs and checked against structured solve-level diagnostics. Controller calls are not dropped on the basis of solver status or gap availability.

6. Additional numerical results

6.1. Primary inference and implementation accounting

As a test-set stability diagnostic, individual restarts had lower mean cost than the better online MIQP in 0/5 at 20 EUR/h, 0/25 at 40 EUR/h, 0/5 at 80 EUR/h; this diagnostic did not alter validation-based selection. At 20 EUR/h, the one-sided upper bound against no storage was -44.6 EUR/day; superiority was established. The zero-action feasible incumbent was retained in 0/1 batches at 20 EUR/h, 0/5 batches at 40 EUR/h, 0/1 batches at 80 EUR/h; retained fallback outcomes are not attributed to neural control. Validation selected the threshold-rule baseline as 20 EUR/h: zero action, 40 EUR/h: zero action, 80 EUR/h: zero action; a zero-action choice remains labeled as such in its tuning log. A complete five-restart batch used 23.5 minutes of recorded training-run time, validation-controller computation, and selected-checkpoint loading. Among trained selections, mean checkpoint loading was 0.02 s. At 96 calls per day, the measured online controller time was 0.30 s/day for the offline policy and 101.27 s/day for conditional-mean MIQP. On compute time alone, assigning the same unit value to offline and online seconds, the measured offline work equals the accumulated online time

Table 3: All prespecified paired comparisons. Differences and bounds are learned minus comparator in EUR/day; fee rates are in EUR/h. The one-sided 95% upper bound uses paired circular date blocks and, at the central fee, complete training batches. Superiority requires a negative upper bound. Holm p adjusts all 15 comparisons.

Fee	Comparator	Difference	Relative	Upper bound	Holm p
20	No storage	-57.0	-4.3%	-44.6	0.001
20	Price rule	-57.0	-4.3%	-44.3	0.001
20	Envelope MPC	-57.0	-4.3%	-44.4	0.001
20	CM MIQP	10.3	0.8%	13.6	1.000
20	Stochastic MIQP	14.7	1.2%	17.4	1.000
40	No storage	-261.5	-14.9%	-241.6	0.001
40	Price rule	-261.5	-14.9%	-241.7	0.001
40	Envelope MPC	-261.4	-14.9%	-241.5	0.001
40	CM MIQP	24.3	1.7%	30.3	1.000
40	Stochastic MIQP	33.1	2.3%	40.2	1.000
80	No storage	-851.6	-32.5%	-823.1	0.001
80	Price rule	-851.6	-32.5%	-822.1	0.001
80	Envelope MPC	-851.6	-32.5%	-822.5	0.001
80	CM MIQP	-12.5	-0.7%	-6.4	0.001
80	Stochastic MIQP	23.6	1.4%	30.1	1.000

saving after 5.3–12.2 deployment days relative to the better of the two online MIQPs. The main paper’s break-even condition also includes the observed operating-cost difference for any chosen monetary value of compute time.

At 20 EUR/h, the learned-minus-reference cost difference was -37.5 EUR/day, with a one-sided upper bound of -31.6 EUR/day; superiority was established. At 40 EUR/h, the learned-minus-reference cost difference was -201.7 EUR/day, with a one-sided upper bound of -192.0 EUR/day; superiority was established. At 80 EUR/h, the learned-minus-reference cost difference was -576.5 EUR/day, with a one-sided upper bound of -549.7 EUR/day; superiority was established.

Table 4: Threshold–fee–smoothing factorial. Δ is offline minus conditional-mean MIQP cost. Threshold and width are in kW, fee in EUR/h, and costs in EUR/day. Near 10 is the number of hours/day within 10 kW of the threshold.

Threshold	Fee	Width	Offline cost	Δ	Smooth–hard	Near 10	Restarts
250	20	5	1286.9	8.1	62.4	6.30	1
250	20	10	1289.5	10.6	58.0	5.77	1
250	20	20	1293.4	14.5	58.7	5.83	1
250	40	5	1566.4	27.5	194.1	9.76	1
250	40	10	1566.4	27.5	192.8	9.65	1
250	40	20	1557.6	18.7	205.1	10.07	1
250	80	5	1943.3	28.9	543.7	13.60	1
250	80	10	1952.6	38.2	527.0	13.19	1
250	80	20	1930.7	16.2	571.2	14.28	1
300	20	5	1255.9	7.9	72.5	7.32	1
300	20	10	1258.2	10.3	68.8	6.87	5
300	20	20	1257.5	9.5	69.8	6.77	1
300	40	5	1493.8	29.0	235.9	11.84	1
300	40	10	1489.1	24.3	225.5	11.25	25
300	40	20	1493.4	28.6	214.7	10.55	1
300	80	5	1800.3	18.1	622.2	15.57	1
300	80	10	1769.7	-12.5	640.0	16.02	5
300	80	20	1772.8	-9.4	644.0	16.09	1
350	20	5	1206.8	-2.1	90.0	9.04	1
350	20	10	1212.6	3.7	78.3	7.83	1
350	20	20	1218.1	9.3	74.7	7.20	1
350	40	5	1413.0	31.2	238.6	12.01	1
350	40	10	1422.5	40.8	241.2	12.04	1
350	40	20	1401.0	19.2	267.9	13.23	1
350	80	5	1622.4	-29.7	685.4	17.15	1
350	80	10	1621.4	-30.7	687.2	17.18	1
350	80	20	1618.6	-33.5	693.6	17.31	1

Table 5: Learned-correction attribution by cost component. Fees are in EUR/h, cost components in EUR/day, throughput in kWh/day, and exceedance in hours/day. The central learned row averages five complete training batches.

Fee	Controller	Bill	Band fee	Degradation	Terminal	Throughput	Exceed h
20	No-learning reference	880.0	412.0	3.6	0.16	97	20.60
20	Offline learned	888.9	304.0	65.1	0.24	1735	15.20
40	No-learning reference	880.2	804.0	6.5	0.19	172	20.10
40	Offline learned	897.1	436.5	155.1	0.37	4136	10.91
80	No-learning reference	881.9	1436.7	27.4	0.24	732	17.96
80	Offline learned	900.5	603.3	265.5	0.42	7080	7.54

Table 6: Winter common-objective decomposition. Cost components are in EUR/day and fee rates in EUR/h.

Fee	Method	Bill	Band fee	Degradation	Terminal	Total
20	No storage	879.9	435.3	0.0	0.00	1315.3
	Validation-tuned price rule	879.9	435.3	0.0	0.00	1315.3
	Convex-envelope MPC	879.7	435.3	0.1	0.07	1315.2
	Conditional-mean MIQP	887.3	266.5	91.7	2.45	1248.0
	Two-stage stochastic MIQP	887.2	261.3	94.2	0.83	1243.5
	Offline procedure	888.9	304.0	65.1	0.24	1258.2
40	No storage	879.9	870.7	0.0	0.00	1750.6
	Validation-tuned price rule	879.9	870.7	0.0	0.00	1750.6
	Convex-envelope MPC	879.7	870.7	0.1	0.10	1750.6
	Conditional-mean MIQP	905.2	350.7	204.7	4.20	1464.8
	Two-stage stochastic MIQP	908.1	318.3	227.9	1.74	1456.0
	Offline procedure	897.1	436.5	155.1	0.37	1489.1
80	No storage	879.9	1741.3	0.0	0.00	2621.3
	Validation-tuned price rule	879.9	1741.3	0.0	0.00	2621.3
	Convex-envelope MPC	879.6	1741.3	0.2	0.20	2621.3
	Conditional-mean MIQP	915.1	638.0	223.4	5.69	1782.2
	Two-stage stochastic MIQP	914.0	583.3	245.3	3.38	1746.1
	Offline procedure	900.5	603.3	265.5	0.42	1769.7

Table 7: Descriptive solver-budget sensitivity at 80 EUR/h. Costs and cost changes are in EUR/day; latency is mean controller-call time. The primary deployment budget is 5 s. Finite gap is the availability fraction; mean gaps are conditional on availability.

Method	Limit	Cost	Cost change	Latency (ms)	Incumbent	Finite gap	Mean gap	Time limit
Conditional-mean MIQP	5 s	1782.2	0.0	1701	100.0%	100.0%	0.66%	10.5%
	15 s	1782.9	0.7	2346	100.0%	100.0%	0.64%	5.5%
Two-stage stochastic MIQP	5 s	1746.1	0.0	2759	100.0%	100.0%	0.84%	27.8%
	15 s	1744.7	-1.4	4497	100.0%	100.0%	0.76%	13.2%

Table 8: Restart and measured offline-computation diagnostics. Fees are in EUR/h; cost ranges and standard deviations are in EUR/day. Offline time is in minutes per five-restart batch.

Fee	Restarts	Batches	Fallbacks	Validation range	Test SD	Selected test range	Min/batch
20	5	1	0	1060.1–1084.2	8.0	1258.2–1258.2	22.8
40	25	5	0	1296.3–1375.1	25.3	1485.1–1501.1	23.7
80	5	1	0	1606.1–1736.1	38.9	1769.7–1769.7	23.3

Table 9: Compute-equivalent amortization against the better of the two online MIQPs. Cost Δ is offline minus comparator in EUR/day; fees are in EUR/h. The last column ignores operating-cost differences and values offline and online compute seconds equally.

Fee	Online comparator	Cost Δ	Saved s/day	Break-even days
20	Two-stage stochastic MIQP	14.7	112.5	12.2
40	Two-stage stochastic MIQP	33.1	185.3	7.7
80	Two-stage stochastic MIQP	23.6	264.6	5.3

Table 10: Winter online-solver diagnostics. Fees are in EUR/h. Gaps are reported only for solves returning finite solver bounds; finite gap is their availability fraction. Solve time excludes model construction. CM and Stoch. denote conditional-mean and two-stage stochastic MIQP; Envelope denotes the convex-tariff-surrogate MPC.

Fee	Method	Solves	Incumbent	Finite gap	Mean gap	P95 solve (s)
20	Envelope MPC	2880	100.0%	100.0%	0.73%	0.11
	CM MIQP	2880	100.0%	100.0%	0.20%	1.16
	Stoch. MIQP	2880	100.0%	100.0%	0.50%	2.96
40	Envelope MPC	2880	100.0%	100.0%	0.66%	0.10
	CM MIQP	2880	100.0%	100.0%	0.27%	2.54
	Stoch. MIQP	2880	100.0%	100.0%	0.39%	4.50
80	Envelope MPC	2880	100.0%	100.0%	0.69%	0.10
	CM MIQP	2880	100.0%	100.0%	0.66%	4.69
	Stoch. MIQP	2880	100.0%	100.0%	0.84%	4.98

Table 11: Seasonal terminal and online-latency diagnostics. Terminal penalties are in EUR/day; latency is mean controller-call time.

Season	Method	Terminal penalty	Mean latency (ms)
Winter	No storage	0.00	0.0
	Validation-tuned price rule	0.00	0.0
	Convex-envelope MPC	0.10	53.9
	Conditional-mean MIQP	4.20	1047.1
	Two-stage stochastic	1.74	1933.3
	MIQP		
	Offline procedure	0.37	3.1
Spring	No storage	0.00	0.0
	Validation-tuned price rule	0.00	0.0
	Convex-envelope MPC	0.09	104.3
	Conditional-mean MIQP	3.43	1112.3
	Two-stage stochastic	2.65	1990.5
	MIQP		
	Offline procedure	0.20	3.1
Summer	No storage	0.00	0.0
	Validation-tuned price rule	0.00	0.0
	Convex-envelope MPC	0.07	78.0
	Conditional-mean MIQP	5.24	908.1
	Two-stage stochastic	2.29	1853.0
	MIQP		
	Offline procedure	0.08	3.0
Autumn	No storage	0.00	0.0
	Validation-tuned price rule	0.00	0.0
	Convex-envelope MPC	0.08	72.6
	Conditional-mean MIQP	3.83	838.4
	Two-stage stochastic	2.53	1873.7
	MIQP		
	Offline procedure	0.00	3.1

Table 12: Scenario-count cost decomposition, averaged over dates and available randomization units. Fees are in EUR/h and cost components in EUR/day.

Fee	M	Limit	Method	Bill	Band fee	Degradation	Terminal	Total
40	–	–	Offline learned	897.1	436.5	155.1	0.37	1489.1
40	2	5 s	Stochastic MIQP	907.9	320.0	226.2	2.06	1456.1
40	8	5 s	Stochastic MIQP	892.7	668.6	117.3	1.70	1680.3
40	16	5 s	Stochastic MIQP	886.6	760.4	60.0	1.11	1708.1
80	–	–	Offline learned	900.5	603.3	265.5	0.42	1769.7
80	16	5 s	Stochastic MIQP	886.6	1521.8	59.2	1.85	2469.5
80	16	15 s	Stochastic MIQP	896.6	1318.2	125.2	2.17	2342.1

Table 13: Scenario-stream and solver diagnostics. Fees are in EUR/h; stream columns are mean costs in EUR/day. Finite gap is the share of calls with finite primal and dual bounds and an ordinary reported gap; the mean gap is conditional on that event. TL is time-limit frequency.

Fee	M	Limit	Stream 1	Stream 2	Stream 3	Incumbent	Finite gap	Mean gap	TL	P95 solve
40	2	5 s	1456.4	1456.0	1455.9	100.0%	100.0%	0.38%	7.8%	4.52 s
40	8	5 s	1682.3	1681.0	1677.5	100.0%	96.4%	821.36%	78.9%	4.98 s
40	16	5 s	1708.1	1707.9	1708.3	100.0%	49.4%	614.81%	96.6%	4.98 s
80	16	5 s	2468.0	2468.5	2471.9	100.0%	48.5%	897.50%	96.8%	4.98 s
80	16	15 s	2340.9	2336.6	2348.8	100.0%	97.7%	1167.00%	96.4%	14.98 s

Table 14: Distribution of paired daily component differences at 40 EUR/h. Entries are learned minus comparator in EUR/day; learned batches and scenario streams are averaged within date.

Comparator	Component	Mean	P10	Median	P90
No-learning reference	Bill	17.0	13.3	17.3	20.7
No-learning reference	Band fee	-367.5	-434.0	-367.0	-310.6
No-learning reference	Degradation	148.6	118.8	147.9	181.9
No-learning reference	Terminal	0.2	-0.1	-0.1	0.1
No-learning reference	Total	-201.7	-231.5	-202.5	-167.6
Stochastic MIQP ($M = 16$)	Bill	10.6	6.6	10.9	15.0
Stochastic MIQP ($M = 16$)	Band fee	-323.9	-385.0	-326.7	-282.2
Stochastic MIQP ($M = 16$)	Degradation	95.1	73.2	93.6	119.2
Stochastic MIQP ($M = 16$)	Terminal	-0.7	-1.8	-0.7	0.0
Stochastic MIQP ($M = 16$)	Total	-218.9	-285.0	-214.8	-163.1

7. Code availability

Code, configuration files, stored weights, and numerical results are available in the GitHub repository at <https://github.com/yeoneung/rvpinn>. The code, numerical results, and accompanying manuscript are provided in version 1.0.0 [6]. The repository includes scripts for regenerating the reported tables. Automated tests cover units, sign convention, SoC safety, OU exact steps, reflection, automatic versus finite-difference derivatives, fixed-policy finite differences, hard tariff action search, one-step terminal behavior, common cost components, and the finite-horizon Bellman bound. Tests also verify the zero-correction reference controller, scenario nesting across M , and the three-level paired resampling calculation.

References

- [1] Open Power System Data, Data package time series, Open Power System Data, 2020. doi:[10.25832/time_series/2020-10-06](https://doi.org/10.25832/time_series/2020-10-06), [dataset], version 2020-10-06.
- [2] B. Xu, A. Oudalov, A. Ulbig, G. Andersson, D. S. Kirschen, Modeling of lithium-ion battery degradation for cell life assessment, *IEEE Transactions on Smart Grid* 9 (2018) 1131–1140. doi:[10.1109/TSG.2016.2578950](https://doi.org/10.1109/TSG.2016.2578950).
- [3] T. Achterberg, SCIP: Solving constraint integer programs, *Mathematical Programming Computation* 1 (2009) 1–41. doi:[10.1007/s12532-008-0001-1](https://doi.org/10.1007/s12532-008-0001-1).
- [4] D. N. Politis, J. P. Romano, A circular block-resampling procedure for stationary data, in: R. LePage, L. Billard (Eds.), *Exploring the Limits of Bootstrap*, Wiley, New York, 1992, pp. 263–270.
- [5] S. Holm, A simple sequentially rejective multiple test procedure, *Scandinavian Journal of Statistics* 6 (1979) 65–70. URL: <https://www.jstor.org/stable/4615733>.
- [6] Y. Kim, Battery dispatch: Code, processed inputs, trained weights, and evaluation outputs, GitHub, 2026. URL: <https://github.com/yeoneung/rvpinn/tree/v1.0.0>, [dataset and software], version 1.0.0, accessed 13 September 2026.